GEOMETRY UPDATE / 06 SEP 2026

INTERNAL TECHNICAL POSTER

TOROIDAL GEOMETRY WHAT CHANGED TODAY

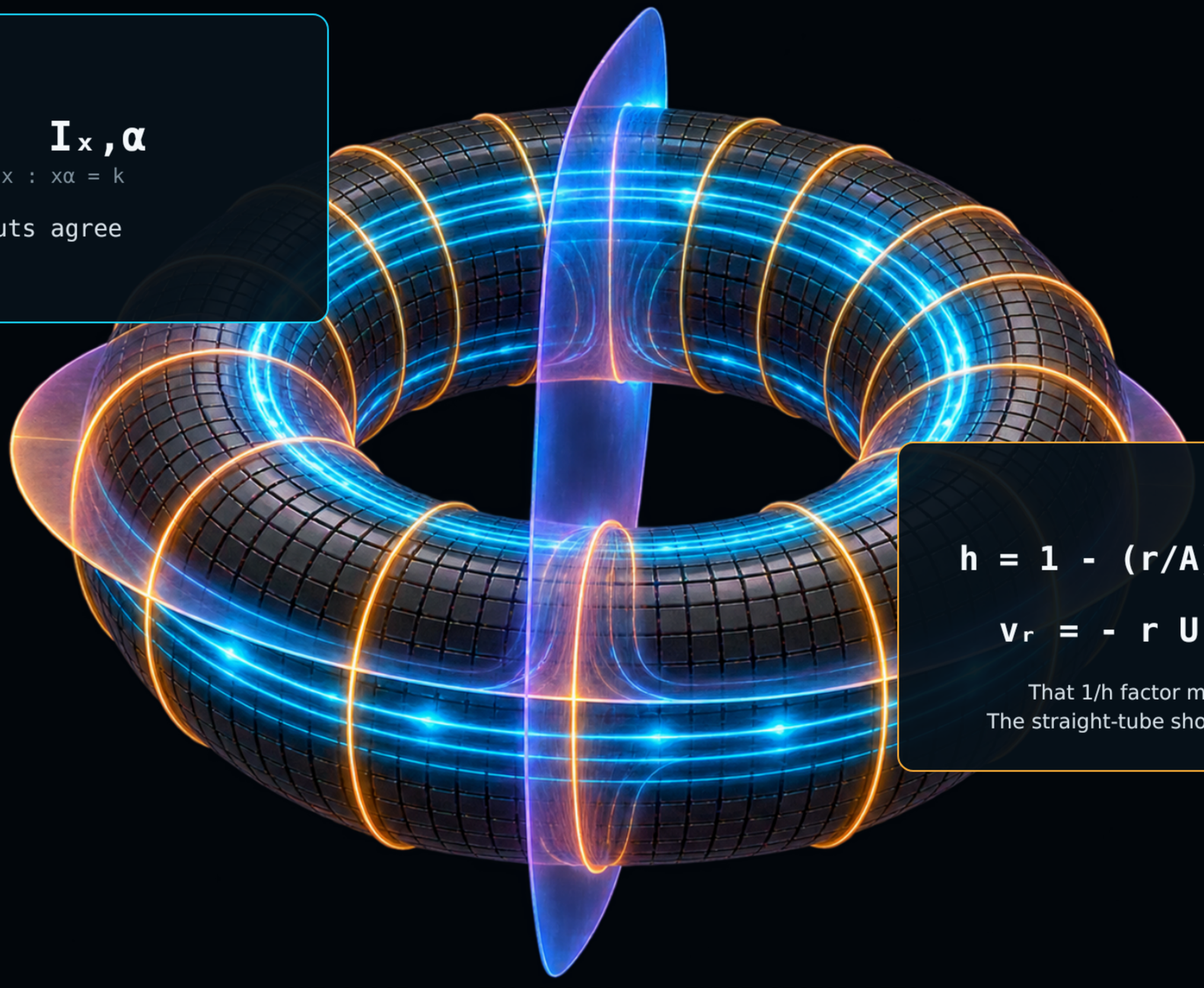
Cut flux is now the winding invariant. Curvature is built into the throat flow.
A fully specified Monte Carlo reference kernel now exists.

CUT FLUX

$$F_\alpha(k) = \sum_{x : x\alpha = k} I_{x,\alpha}$$

$\operatorname{div} I = 0 \Rightarrow$ cuts agree

$W_\alpha = F_\alpha(k) \in \mathbb{Z}$



CURVED THROAT

$$h = 1 - (r/A)\cos\theta > 0$$

$$v_r = -\,r\,U'(s)\,/\,(2h)$$

That 1/h factor makes $\nabla\cdot v = 0$ exactly.
The straight-tube shortcut fails in general.

ONE SPACE • TWO CYCLES • THREE SECTOR BITS

RECTANGULAR TORUS

DIAGONAL ROUTING

Subgroup index = gcd(2, M, N).
Two sheets iff M and N are both even.

If either period is odd, (x+y) mod 2 is not globally defined.

REFERENCE DYNAMICS

A REAL KERNEL

X = (I, M, q) on a 2×2×2 torus.
71 weighted microticks per sweep.

Symmetric proposals + Metropolis give detailed balance; full-state reachability and identity close the kernel proof.

PHASE BRIDGE **13 → 39 coverings are explicit. 39-point quadrature is exact for input modes |m| ≤ 12 - not globally.**

PROVED

cut = winding
curved ∇·v = 0
detailed balance
reachability + aperiodicity

EXECUTABLE

chain-complex validator
64 routing rectangles
modular current checks
kernel contract audit

NOT YET RUN

production sampler
warmup / mixing
Q2 / equilibrium
empirical throat test

NO MYSTICISM • EXACT MAPS • EXECUTION STAYS SEPARATE

SOURCES Toroidal Geometry v1.0 • Toroidal Dynamics v0.1 • Geometry Maximization v1.61 + v1.6 receipt